

Group 2

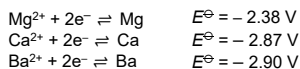
Trends: Physical properties

Properties	Trends
atomic radii/ ionic radii	_____ down the group - shielding effect (no. of e⁻ shell) - nuclear charge - electrostatic attraction between nucleus and <u>outermost e⁻</u> ⇒ e⁻ cloud size
1 st I.E.	_____ down the group - shielding effect (no. of e⁻ shell) - nuclear charge - electrostatic attraction between nucleus and <u>outermost e⁻</u> ⇒ energy req. to remove outermost e⁻
electronegativity	_____ down the group - shielding effect (no. of e⁻ shell) - nuclear charge - electrostatic attraction between nucleus and <u>bonding e⁻</u>

Trends: Chemical properties

Properties	Trends
reactivity of metal (M)	_____ down the group

E° value from Data Booklet:



How does trend in E° value explain reactivity?

Down the group,

- E° value becomes more _____
- tendency for _____
reaction increases
- tendency of metal losing e⁻ _____
- reducing power of metal _____
- reactivity of metal _____

Carbonates: Thermal stability

Properties	Trends
decomposition temperature	$\text{MCO}_3 \rightarrow \text{MO} + \text{CO}_2$ increases down the group

illustration:



Explain why does thermal stability of Group 2 carbonates increase down the group.

Down the group,

- cationic radii _____
- charge density of cation _____
- polarising power of cation _____
- extent of distortion on electron cloud of CO₃²⁻ _____
- extent of weakening of covalent bond in CO₃²⁻ _____
- energy req to break the covalent bonds _____

Group 17

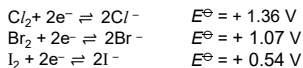
Trends: Physical properties

Properties	Trends
atomic radii/ ionic radii	_____ down the group
1 st I.E.	_____ down the group
electronegativity	_____ down the group
bond dissociation energy of X ₂	_____ down the group - diffuse-ness of valence orbital - effectiveness of orbital overlap
volatility of X ₂	_____ down the group - size and polarisability of electron cloud - strength of id-id - energy req to overcome

Trends: Chemical properties

Properties	Trends
reactivity of halogen (X ₂)	_____ down the group

E° value from Data Booklet:



How does trend in E° value explain reactivity?

Down the group,

- E° value becomes less _____
- tendency for _____
reaction decreases
- tendency of halogen gaining e⁻ _____
- oxidising power of halogen _____
- reactivity of halogen _____

HX: Thermal stability

Properties	Trends
decomposition temperature	$\text{HX} \rightarrow \frac{1}{2} \text{H}_2 + \frac{1}{2} \text{X}_2$ decreases down the group

Note: HF and HCl do not decompose on heating

Explain why does thermal stability of hydrogen halides decrease down the group.

Down the group,

- the valence p orbital of the halogen becomes _____.
Hence there is _____ overlap between the 1s orbital of hydrogen and the valence p orbital of halogen.
- the electronegativity difference between hydrogen and halogen _____, resulting in a _____ bond.

Hence, H-X bond strength _____ down the group

Displacement reactions

	Cl ⁻	Br ⁻	I ⁻
Cl ₂	—		
Br ₂		—	
I ₂			—

Notes:

- more reactive halogen displaces less reactive halide

Reaction with thiosulfate

Cl ₂ & Br ₂	Oxidises S ₂ O ₃ ²⁻ to _____
I ₂	Oxidises S ₂ O ₃ ²⁻ to _____

Write half eqⁿ then combine:

[O]:
[R]:
Overall :

Reaction with Fe

Cl ₂ & Br ₂	Oxidises Fe to _____
I ₂	Oxidises Fe to _____

Write half eqⁿ then combine:

[O]:
[R]:
Overall :

Test for X⁻

Test	Observation & identity
Step 1:	For Cl ⁻ :
Step 2:	For Br ⁻ :
	For I ⁻ :

Group 2

Trends: Physical properties

Properties	Trends
atomic radii/ ionic radii	<u>increases</u> down the group ① shielding effect (no. of e ⁻ shell) ② nuclear charge ③ electrostatic attraction between nucleus and <u>outermost e⁻</u> ⇒ e ⁻ cloud size
1 st I.E.	<u>decreases</u> down the group ① shielding effect (no. of e ⁻ shell) ② nuclear charge ③ electrostatic attraction between nucleus and <u>outermost e⁻</u> ⇒ energy req. to remove outermost e ⁻
electronegativity	<u>decreases</u> down the group ① shielding effect (no. of e ⁻ shell) ② nuclear charge ③ electrostatic attraction between nucleus and <u>bonding e⁻</u>

Trends: Chemical properties

Properties	Trends
reactivity of metal (M)	<u>increases</u> down the group

E^o value from Data Booklet:

$$\begin{array}{ll} \text{Mg}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Mg} & E^{\circ} = -2.38 \text{ V} \\ \text{Ca}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Ca} & E^{\circ} = -2.87 \text{ V} \\ \text{Ba}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Ba} & E^{\circ} = -2.90 \text{ V} \end{array}$$

How does trend in E^o value explain reactivity?

Down the group,

- E^o value becomes more negative
- tendency for backward oxidation reaction increases
- tendency of metal losing e⁻ increases
- reducing power of metal increases
- reactivity of metal increases

Carbonates: Thermal stability

Properties	Trends
decomposition temperature	$\text{MCO}_3 \rightarrow \text{MO} + \text{CO}_2$ <u>increases</u> down the group

illustration:

Explain why does thermal stability of Group 2 carbonates increase down the group.

Down the group,

- cationic radii increases
- charge density of cation decreases
- polarising power of cation decreases
- extent of distortion on electron cloud of CO₃²⁻ decreases
- extent of weakening of covalent bond in CO₃²⁻ decreases
- energy req to break the covalent bonds increases

Group 17

Trends: Physical properties

Properties	Trends
atomic radii/ ionic radii	<u>increases</u> down the group
1 st I.E.	<u>decreases</u> down the group
electronegativity	<u>decreases</u> down the group
bond dissociation energy of X ₂	<u>decreases</u> down the group • diffuse-ness of valence orbital • effectiveness of orbital overlap
volatility of X ₂	<u>decreases</u> down the group • size and polarisability of electron cloud • strength of id-id • energy req to overcome <u>intermolecular force of attraction</u>

Trends: Chemical properties

Properties	Trends
reactivity of halogen (X ₂)	<u>decreases</u> down the group

E^o value from Data Booklet:

$$\begin{array}{ll} \text{Cl}_2 + 2\text{e}^{-} \rightleftharpoons 2\text{Cl}^{-} & E^{\circ} = +1.36 \text{ V} \\ \text{Br}_2 + 2\text{e}^{-} \rightleftharpoons 2\text{Br}^{-} & E^{\circ} = +1.07 \text{ V} \\ \text{I}_2 + 2\text{e}^{-} \rightleftharpoons 2\text{I}^{-} & E^{\circ} = +0.54 \text{ V} \end{array}$$

How does trend in E^o value explain reactivity?

Down the group,

- E^o value becomes less positive
- tendency for forward reduction reaction decreases
- tendency of halogen gaining e⁻ decreases
- oxidising power of halogen decreases
- reactivity of halogen decreases

HX: Thermal stability

Properties	Trends
decomposition temperature	$\text{HX} \rightarrow \frac{1}{2} \text{H}_2 + \frac{1}{2} \text{X}_2$ <u>decreases</u> down the group

Note: HF and HCl do not decompose on heating

Explain why does thermal stability of hydrogen halides decrease down the group.

Down the group,

- the valence p orbital of the halogen becomes more diffuse. Hence there is less effective overlap between the 1s orbital of hydrogen and the valence p orbital of halogen.
- the electronegativity difference between hydrogen and halogen decreases, resulting in a less polar bond.

Hence, H-X bond strength decreases down the group

Displacement reactions

	Cl ⁻	Br ⁻	I ⁻
Cl ₂	—	$\text{Cl}_2 + 2\text{Br}^{-} \rightarrow 2\text{Cl}^{-} + \text{Br}_2$	$\text{Cl}_2 + 2\text{I}^{-} \rightarrow 2\text{Cl}^{-} + \text{I}_2$
Br ₂	no reaction	—	$\text{Br}_2 + 2\text{I}^{-} \rightarrow 2\text{Br}^{-} + \text{I}_2$
I ₂	no reaction	no reaction	—

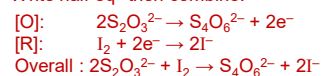
Notes:

- more reactive halogen displaces less reactive halide

Reaction with thiosulfate

Cl ₂ & Br ₂	Oxidises S ₂ O ₃ ²⁻ to <u>SO₄²⁻</u>
I ₂	Oxidises S ₂ O ₃ ²⁻ to <u>S₄O₆²⁻</u>

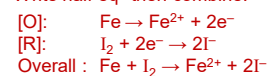
Write half eqⁿ then combine:



Reaction with Fe

Cl ₂ & Br ₂	Oxidises Fe to <u>Fe³⁺</u>
I ₂	Oxidises Fe to <u>Fe²⁺</u>

Write half eqⁿ then combine:



Test for X⁻

Test	Observation & identity
Step 1: add AgNO ₃ (aq)	For Cl ⁻ : <u>white ppt of AgCl formed, soluble in excess NH₃(aq)</u>
Step 2: add excess NH ₃ (aq) OR add conc. NH ₃	For Br ⁻ : <u>cream ppt of AgBr formed, soluble in conc NH₃</u>
	For I ⁻ : <u>yellow ppt of AgI formed, insoluble in conc NH₃</u>